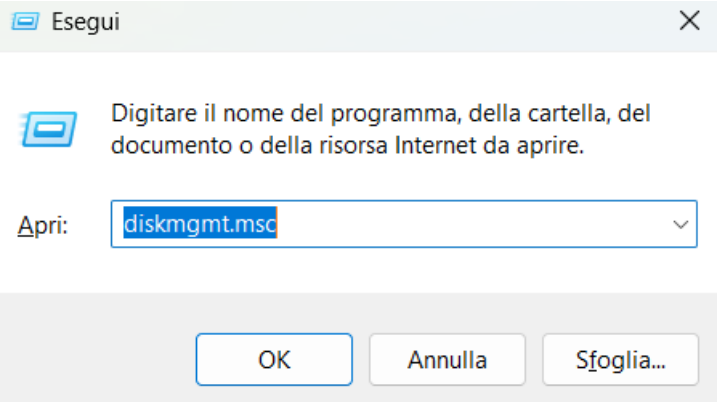
Para-virtualized Dom-U on Xen

Di seguito gli step che ho eseguito per avviare una macchina virtuale paravirtualizzata su Xen hypervisor come Dom-U.

Creazione Partizione LVM

Da un PC Windows-native ho creato una partizione raw:

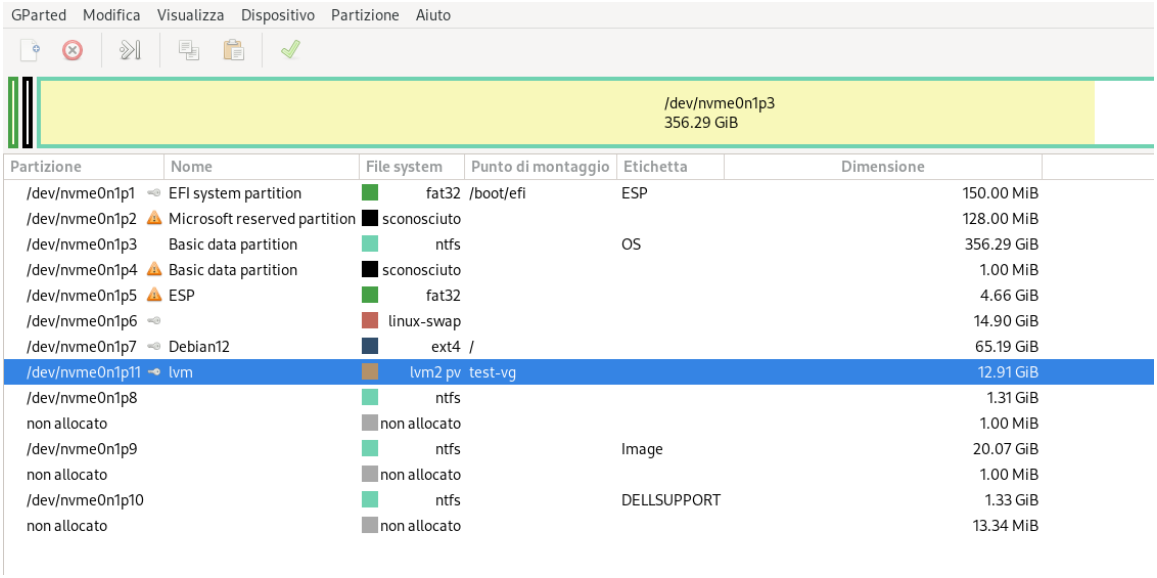
- Win + R , digitare diskmgmt.msc e cliccare 'OK':



- Individuare una partizione di memoria (della dimensione desiderata) non allocata. Cliccare con il tasto destro -> Nuovo volume semplice.
  Non selezionare alcun tipo di filesystem. Verrà creata una partizione RAW.

Volume	Layout	Tipo	File system
(D:)	Semplice	Di base	RAW

- Riavviare il PC e da GRUB accedere alla partizione Linux (nel mio caso distro Debian-12).
- Aprire GParted e formattare con file system LVM la partizione RAW precedentemente creata:



Partizione	Nome	File system	Punto di montaggio	Etichetta	Dimensione
/dev/nvme0n1p1	EFI system partition	fat32	/boot/efi	ESP	150.00 MiB
/dev/nvme0n1p2	Microsoft reserved partition	sconosciuto			128.00 MiB
/dev/nvme0n1p3	Basic data partition	ntfs		OS	356.29 GiB
/dev/nvme0n1p4	Basic data partition	sconosciuto			1.00 MiB
/dev/nvme0n1p5	ESP	fat32			4.66 GiB
/dev/nvme0n1p6		linux-swap			14.90 GiB
/dev/nvme0n1p7	Debian12	ext4	/		65.19 GiB
/dev/nvme0n1p11	lvm	lvm2 pv	test-vg		12.91 GiB
/dev/nvme0n1p8		ntfs			1.31 GiB
non allocato		non allocato			1.00 MiB
/dev/nvme0n1p9		ntfs		Image	20.07 GiB
non allocato		non allocato			1.00 MiB
/dev/nvme0n1p10		ntfs		DELLSUPPORT	1.33 GiB
non allocato		non allocato			13.34 MiB

- Da questo punto in poi sono stati seguiti i passaggi riportati ai seguenti link(verranno ugualmente riportati sotto):
  1- virtualization_technologies_course/2_paravirtualization at master · ldesi/virtualization_technologies_course · GitHub .
  2- Xen Project Beginners Guide - Xen .

Setup Network

1- # apt-get install bridge-utils.
2- Ho collegato il PC tramite cavo LAN al modem.
3- # ip a

```
andrea@AndreaPC:~$ su -
Password:
root@AndreaPC:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp59s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether b0:7b:25:65:9a:c5 brd ff:ff:ff:ff:ff:ff
3: wlp60s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether a4:97:b1:c5:09:01 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.25/24 brd 192.168.1.255 scope global dynamic noprefixroute wlp60s0
       valid_lft 86400sec preferred_lft 86400sec
    inet6 fe80::a697:b1ff:fec5:901/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@AndreaPC:~#
```

4- # sudo nano /etc/netwrok/interfaces

```
  GNU nano 7.2
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback


allow-hotplug enp59s0
iface enp59s0 inet manual

auto xenbr0
iface xenbr0 inet dhcp
        bridge_ports enp59s0
```

5- Per verificare che il bridge xenbr0 sia attivo → # brctl show

LVM

1- # pvcreate /dev/nvmeXXX
   Dove nvmeXXX il disco nel quale è stato creato il LVM. Nel mio caso: /dev/nvme0n1p11.
2- # vgcreate test-vg /dev/nvme0n1p11 .
3- # lvcreate -L 5G -n lv_vm_ubuntu /dev/test-vg .
4- Scaricare le immagini netboot:
   # wget http://archive.ubuntu.com/ubuntu/dists/bionic-updates/main/installer-amd64/current/images/netboot/xen/vmlinuz
   # wget http://archive.ubuntu.com/ubuntu/dists/bionic-updates/main/installer-amd64/current/images/netboot/xen/initrd.gz
5- Creazione del file di configurazione “ubuntu-pv.cfg”:

```
kernel = "/root/vmlinuz"

ramdisk = "/root/initrd.gz"

memory = 1024

vcpus = 2

vif = ['bridge=xenbr0']

disk = [ '/dev/test-vg/lv_vm_ubuntu,raw,xvda,rw' ]

#bootloader = "pygrub"
```

Una volta completata l'installazione, commentare le righe relative al kernel e alla ramdisk e decommentare il bootloader.
Nelle versioni più recenti di Xen, non è necessario inserire tutto il path di pygrub.

Per lanciare la vm:
xl create -c ubuntu-pv.cfg

Seguire tutti i passi dell'installazione.

```
┤ [!!] Partition disks ├

If you continue, the changes listed below will be written to the
disks. Otherwise, you will be able to make further changes manually.

The partition tables of the following devices are changed:
   Virtual disk 1 (xvda)

The following partitions are going to be formatted:
   partition #1 of Virtual disk 1 (xvda) as ext4

Write the changes to disks?

    <Yes>                                                  <No>


<Tab> moves; <Space> selects; <Enter> activates buttons
```

Scegliere di installare il bootloader GRUB:

```
┤ [!] Install the GRUB boot loader on a hard disk ├

It seems that this new installation is the only operating system on
this computer. If so, it should be safe to install the GRUB boot
loader to the master boot record of your first hard drive.

Warning: If the installer failed to detect another operating system
that is present on your computer, modifying the master boot record
will make that operating system temporarily unbootable, though GRUB
can be manually configured later to boot it.

Install the GRUB boot loader to the master boot record?

    <Go Back>                                    <Yes>    <No>


<Tab> moves; <Space> selects; <Enter> activates buttons
```

Al termine dell'installazione:

```
┤ Finishing the installation ├

95%

The system is going down NOW!ystem...
Sent SIGTERM to all processes
Sent SIGKILL to all processes
Requesting system reboot
[ 2153.603185] reboot: Restarting system
root@AndreaPC:~# xl list
Name                                        ID   Mem VCPUs      State   Time(s)
Domain-0                                     0  6830    12     r-----    1300.6
ubuntu-pv-example                            2  1024     2     -b----       2.0
root@AndreaPC:~# xl shutdown ubuntu-pv-example
Shutting down domain 2
root@AndreaPC:~# xl list
Name                                        ID   Mem VCPUs      State   Time(s)
Domain-0                                     0  6830    12     r-----    1303.8
root@AndreaPC:~#
```